

QUESTION 1 — BASIC PROBABILITY

What is Probability?

Probability is a branch of mathematics that measures the chance of an event occurring. It helps us predict how likely something is to happen. The value of probability always lies between 0 and 1.

- Probability = 0 means the event is impossible.
- Probability = 1 means the event is certain.

For example:

- Probability of getting number 7 on a standard dice is impossible, so probability is 0.
- Probability of getting a number less than 7 on a dice is certain, so probability is 1.

In this project, probability is used to analyze the chances of students passing the exam based on attendance, study hours, previous test scores, and participation in group discussions.

Important Terms in Probability

1. Experiment

An experiment is an activity or process that produces outcomes.

Example:

Checking whether a student passes or fails the exam.

2. Outcome

An outcome is the result of an experiment.

Example:

A student may either pass or fail.

3. Event

An event is a collection of one or more outcomes.

Example:

All students who pass the final exam.

4. Sample Space

A sample space is the complete set of all possible outcomes.

Example:
For exam results:
 $S = \{\text{Pass, Fail}\}$

Examples of Probability from Dataset

1. Probability that a student passes the exam.
 2. Probability that attendance is greater than 80%.
 3. Probability that a student participates in group discussion.
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QUESTION 2 — TYPES OF PROBABILITY

Empirical Probability

Empirical probability is the probability calculated using actual observed data collected from experiments or real-life situations.

It is calculated using the formula:

$$P(E) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Outcomes}}$$

In this project, empirical probability is calculated using the dataset of students.

Example:
If 300 students pass out of 500 students, then:

$$P(\text{Pass}) = 300 / 500 = 0.6$$

This means there is a 60% chance of students passing.

Theoretical Probability

Theoretical probability is calculated using mathematical assumptions without using actual data.

Formula:

$$P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Possible Outcomes}}$$

Example:
When tossing a fair coin:

$$P(\text{Head}) = 1/2$$

because there is 1 favorable outcome and 2 possible outcomes.

Difference Between Empirical and Theoretical Probability

Empirical Probability Theoretical Probability

Based on actual data Based on assumptions

Uses observations Uses mathematical calculations

Practical approach Ideal approach

QUESTION 3 — RANDOM VARIABLE AND PROBABILITY DISTRIBUTION

Random Variable

A random variable is a variable whose value depends on the outcomes of a random experiment.

In this project:

X = Number of students passing among 3 randomly selected students

Possible values of X are:

$X = 0, 1, 2, 3$

Where:

- $X = 0$ means no student passes.
 - $X = 1$ means one student passes.
 - $X = 2$ means two students pass.
 - $X = 3$ means all students pass.
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Probability Distribution

A probability distribution shows all possible values of a random variable and their corresponding probabilities.

It helps in understanding how probabilities are distributed among different outcomes.

Binomial Distribution

Binomial distribution is used when:

- there are fixed number of trials,

- only two outcomes are possible,
- trials are independent.

In this project:

- Pass and Fail are two outcomes.
 - Number of selected students is fixed at 3.
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Mean of Binomial Distribution

Mean represents the average expected value.

Formula:

$$\mu = np$$

Where:

- n = number of trials
 - p = probability of success
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Variance of Binomial Distribution

Variance measures how much the values are spread from the mean.

Formula:

$$\sigma^2 = np(1 - p)$$

QUESTION 4 — VENN DIAGRAM

Venn Diagram

A Venn diagram is a diagram used to represent relationships between sets. It uses overlapping circles to show common elements.

In this project:

- Set A represents students with study hours greater than 10.
- Set B represents students with attendance greater than 80%.

The overlapping region shows students satisfying both conditions.

Intersection of Sets

Intersection means common elements between sets.

Formula:

$$A \cap B$$

Where:

- A = Study hours > 10
- B = Attendance $> 80\%$

The intersection contains students satisfying both conditions.

Importance of Venn Diagram

Venn diagrams help in:

- understanding relationships between groups,
 - identifying common elements,
 - visualizing set operations.
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QUESTION 5 — CONTINGENCY TABLE AND PROBABILITIES

Contingency Table

A contingency table is used to show the relationship between two categorical variables.

In this project:

- Group Discussion
- Final Exam Pass

The table helps analyze whether group discussion affects exam performance.

Joint Probability

Joint probability is the probability of two events occurring together.

Example:

Probability that a student participates in group discussion and passes the exam.

Marginal Probability

Marginal probability is the probability of a single event occurring independently.

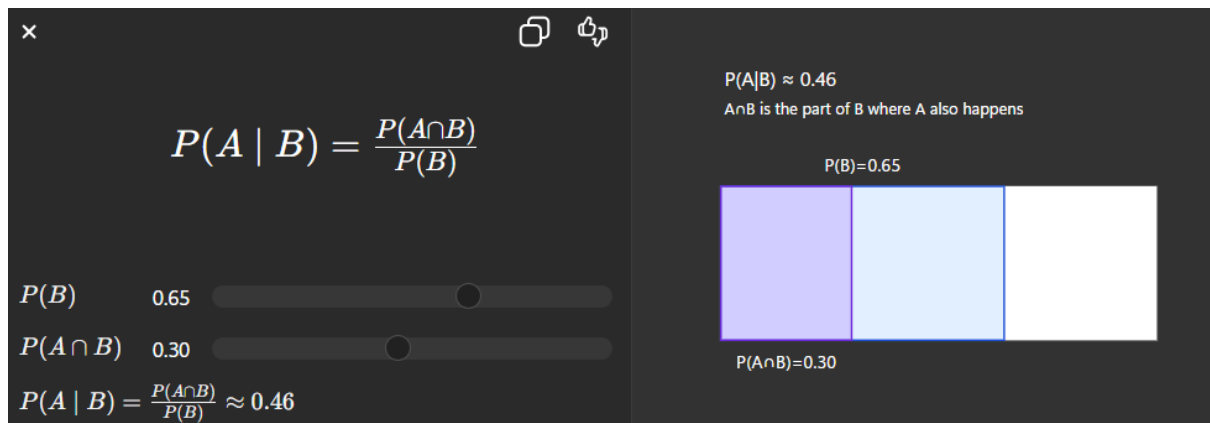
Example:

Probability of students passing the exam.

Conditional Probability

Conditional probability is the probability of one event occurring given that another event has already occurred.

Formula:



Where:

- A = student passes exam
- B = student participates in group discussion

This formula helps determine how one condition affects another.

QUESTION 6 — TYPES OF EVENTS

Independent Events

Two events are independent if occurrence of one event does not affect the occurrence of another event.

Example:

Tossing two coins.

Dependent Events

Two events are dependent if one event affects the occurrence of another event.

Example:

Study hours affecting exam performance.

In this project, attendance and study hours may affect passing probability, so they are dependent events.

Mutually Exclusive Events

Two events are mutually exclusive if they cannot occur together at the same time.

Example:

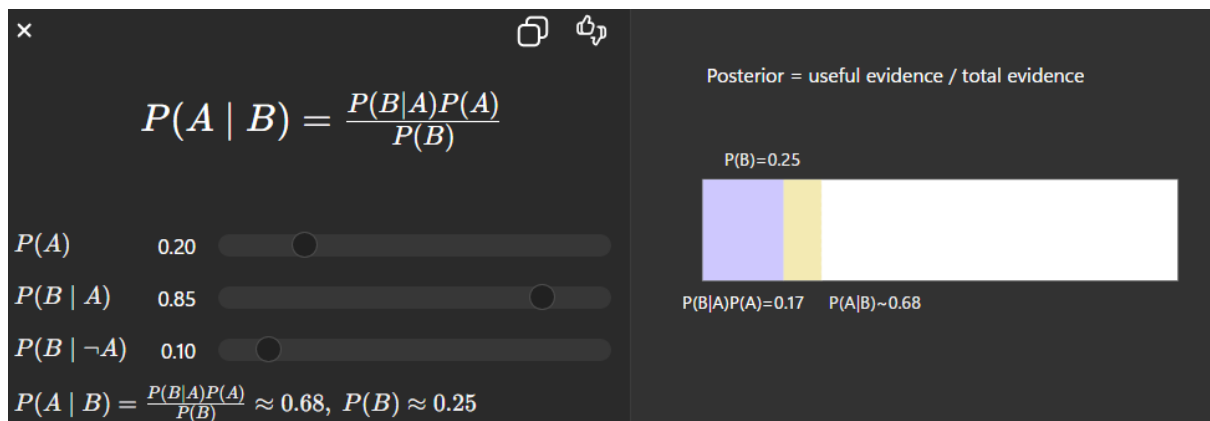
A student cannot both pass and fail simultaneously.

QUESTION 7 — BAYES THEOREM

Bayes Theorem

Bayes theorem is used to calculate conditional probability. It helps determine the probability of an event based on prior information.

Formula:



Where:

- A = student passes exam
- B = student has high attendance

Bayes theorem helps in predicting exam success based on attendance.

Importance of Bayes Theorem

Bayes theorem is useful in:

- prediction,
- machine learning,
- decision making,

- probability analysis.
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FINAL CONCLUSION

This project analyzed student exam performance using concepts of probability and statistics. The dataset was used to calculate probabilities, create distributions, analyze relationships, and visualize data.

The analysis shows that:

- students with higher attendance have greater chances of passing,
- students participating in group discussions perform better,
- study hours positively affect exam performance,
- previous test scores also influence final exam results.

Therefore, attendance, active participation, and regular study habits significantly improve student success and academic performance.